Computer Science **Lab C – POSTFIX PREFIX SOLVER**

Lab Goal :

This lab was designed to teach you more about stacks.

Lab Description :

Take a postfix expression and solve it. All numbers will be in the range 0 to 9.

Infix expression 7 + 8 - 2
Prefix expression - + 7 8 2
Postfix expression 7 8 + 2 -

algorithm help

```
while there are more values in the original expression
{
  get the next value
  if the value is a number
      push the number onto the stack
  else if the value is an operator
      pop 2 numbers from the stack
      use the operator to evaluate the 2 numbers
      push the resulting number onto the stack
}
return the top value from the stack
```

Sample Data :

```
2 7 + 1 2 + +
1 2 3 4 + + -
9 3 * 8 / 4 +
19 3 / 2 * 7 9 * + 43 -
35 5 + 2 * 4 / 19 +
2.3 49.2 + 0.27 * 9 2 / +
2 3 2 ^ ^
```

Sample Output :

```
Evaluating Postfix Expressions:
2 7 + 1 2 + + == 12.0
1 2 3 4 + + - == -8.0
9 3 * 8 / 4 + == 7.375
19 3 / 2 * 7 9 * + 43 - == 32.66666666666667
35 5 + 2 * 4 / 19 + == 39.0
2.3 49.2 + 0.27 * 9 2 / + == 18.405
2 3 2 ^ ^ == 512.0

Evaluating Prefix Expressions:
+ + 2 7 + 1 2 == 12.0
- 1 + 2 + 3 4 == -8.0
+ / * 9 3 8 4 == 7.375
- + * / 19 3 2 * 7 9 43 == 32.66666666666667
+ / * + 35 5 2 4 19 == 39.0
+ * + 2.3 49.2 0.27 / 9 2 == 18.405
^ 2 ^ 3 2 == 512.0
```